

Project

Total Points (pts) =100

The goal of this assignment is to give you hands-on practice with 3D LiDAR data.

1. Submission guidelines

- Create a directory and name it *PJ_<First_Name>_<Last_Name>_<RIT_ID>*. For example, *PJ_Kaleem_Khan_12345678*
- Put all your assignment related files in this directory/folder.
- Compress the directory in a format .zip or equivalent and then submit the compressed file.
- The directory must contain the following files/scripts.
 - Your well-commented file *.py file.
 - **DO NOT** include the image file on which you run the functions.
 - A PDF named *PJ_<First_Name>_<Last_Name>_<RIT_ID>.pdf*, which includes your general ideas to implement your project.

2. Download Data

The challenge is to detect cars at an intersection using lidar data. We will simplify by using a custom synthetic dataset which simulates a lidar sensor mounted above an intersection with simulated cars driving by. You can download the data [here](#).

Also be sure to pip install everything in the requirements file to run the 3D visualization tools.

3. Design, and Implementation of a Perception Stack for Autonomous Driving



Figure 1: Still image from the traffic simulation

In this project, you will design, and implement a perception stack for autonomous driving vehicles. To do this, we will give you a set of 3D point clouds that we have captured from a LiDAR mounted at an intersection. From these point clouds, at each frame, you are supposed to extract 3D points belonging to vehicles. From those points, you will estimate, for each vehicle, its position, bounding box, and motion vector. The bounding box must be a tight-fitting bounding box around a vehicle's 3D points. However, we will only evaluate the **positions of cars and their motion vectors**. **Bounding boxes will not be considered** when calculating accuracy.

CSCI.431 - Intro to Computer Vision

The position of the vehicle is defined by the center point of the vehicle. The motion vector is 3D vector that gives the instantaneous velocity of the vehicle. You can calculate this by subtracting the vehicle's position in the previous frame from the current frame.

Design Goals

The two design goals for autonomous driving are accuracy, and latency. Accuracy (self-explanatory) is how accurately you can estimate a vehicle's position, its bounding box, and its motion vector. We want high accuracy. Latency represents how long it takes your perception stack to process a single frame. We want latency to be low.

Resources

Because you will be working with 3D data, we recommend that you use the Open3D library in your implementation. Open3D has a Python implementation which should significantly speed up your development.

You can find the dataset that you will work with **here**. Please download the dataset. Make a directory **data** in root directory of this assignment. Inside that directory, you should unzip the dataset file. After unzipping the path to the point clouds should be: **data/3d-data/PointClouds/**. Getting this right is important.

Grading Rubric

We have provided a provisional **grading_script.py** file which you can use to test your outputs known ground-truth vehicle locations. We also have held-out test data on our end, so don't over-optimize for the data you've been given.

1. Accuracy (80 marks)

Accuracy is divided into the following subcategories:

- Vehicle position: 45 marks (25 for provided dataset + 20 for held-out test dataset)
- Motion vectors: 35 marks (20 for provided dataset + 15 for held-out test dataset)

It is OK to not have an 80/80 grade for the second part of the assignment. Practically, it is not possible to get 80/80 because in the initial frames, you cannot see the vehicles. The answer key gets a grade of around 60/80 and so we will consider that as 80/80.

Also, we have an unseen dataset that we will evaluate your code on. This consists of approximately 150 point clouds. The idea is to discourage you to get your code to memorize the ground truth. The setting for this dataset is the same as the one we have given you.

2. Runtime Latency (20 marks)

You can check your runtime latency by running your code on one of the servers listed below:

- weasley.cs.rit.edu
- lovegood.cs.rit.edu
- granger.cs.rit.edu

Latency Scoring Criteria:

You will only be graded for latency if you get at least 50% marks for accuracy.

CSCI431 - Intro to Computer Vision

- < 0.1 s per frame: 20 marks
- 0.1 – 0.2 s per frame: 15 marks
- 0.2 – 0.3 s per frame: 10 marks

Please run latency evaluations on the VMs you have been given access to.

Accessing the Servers:

To connect via SSH, use the following command in your terminal:

```
ssh username@weasley.cs.rit.edu
```

Replace username with your RIT username.

You can use the scp command to transfer files to the server. (Please look up online resources for how to use the scp command.)

Output Format

We require a separate CSV file per frame with a header row and a row per detected vehicle giving ID, location, velocity, and bounding box information. Inside of `perception_3d.py`, writing the CSV files in the expected format has essentially been handled for you. Your job is to populate the `vehicles` list each frame.

Important: to simulate real-time processing, your code must output a CSV file in each loop iteration. You may not first loop over all frames doing calculations then later do an output loop.

What is Provided to You

We have provided a script with some boilerplate code for you to get started with (`src/perception_3d.py`). This will contain the code you develop to process the 3D point clouds. It reads point clouds from `data/3d-data/PointClouds/` one-by-one.

`src/3d_demo.py` could visualize one input point cloud at a time.

In addition, you can find the file containing ground truth positions, bounding boxes and motion vectors in `src/ground_truth.csv`.

Collaboration and Generative AI disclosure and Assignment Reflection

Did you collaborate with anyone? Did you use any Generative AI tools? How did the assignment go? Briefly explain what you did in the `collaboration-disclosure.txt` file. **Completing the collaboration disclosure for each assignment is required.** Completing the reflection part of the file is optional, but it is a good way to give feedback to the instructor and grader about how the course is going.

